

Analysis and Compensation of Voltage Distortion by Zero Current Clamping in Voltage-Fed PWM Inverter

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In voltage-fed PWM inverter, the relation between the reference voltage and the output voltage is nonlinear. Especially, when the currents are around zero point, the nonlinear voltage distortion invokes the most serious problems in the system performance. In this paper, the analysis of the voltage distortion by the zero current clamping phenomenon is discussed. From this analysis, a novel distortion voltage compensation strategy that eliminates zero current clamping is presented. Experimental results are also presented to demonstrate the validity of the proposed method.

Key word : Zero Current Clamping, Dead Time

1. Introduction

Voltage-fed PWM inverters have gained increasing popularity in industrial applications. In their application areas, ac machine drive is one of the most important parts. Considering the development of the fast switching devices and modern control theory, a high performance ac machine drive is expected to draw growing attention. In some applications such as sensorless vector control and direct vector control, the inverter output voltages are needed to calculate the desired state variables, especially flux and back-emf. Unfortunately, it is very difficult to measure the output voltage and the additional hardware is required for the measurement. The most desirable and simple method to obtain the output voltage is to use the reference voltage as the output voltage. Generally, the relation between the reference voltage and output voltage has a nonlinear characteristic due to the distorted voltage generated by voltage-fed PWM inverter. Thus unless the distorted voltage is properly compensated, the reference voltage can't be used as the output voltage. For this reason, the accurate output voltage synthesis in voltage-fed PWM inverter is very important for high performance drive system.

In voltage-fed PWM inverter, there are several factors to distort output voltage. Some are originated from the inherent characteristic of the switching devices such as the voltage drop, turn-on and turn-off time and dead time⁽¹⁾. These effects are varying with the direction of the current and should be taken into account for accurate output voltage synthesis. Especially, when the current is nearly zero, the distortion of the output voltage is more severe and the zero current clamping phenomenon results in the deterioration of current waveform. This phenomenon is conspicuous when both the output frequency and voltage are low. Recently remarkable efforts have been made to compensate for the voltage distortion by the dead time⁽¹⁻⁶⁾. However the zero

current clamping phenomenon has hardly been studied and cannot be avoided in most papers⁽¹⁻⁴⁾. The main cause for the zero current clamping phenomenon is a parasitic capacitor parallel to the switch⁽⁵⁻⁶⁾. Charging the parasitic capacitor makes the voltage transition very slow in the case of low current. This effect is clear when the parasitic capacitance is large. The compensation for this effect is difficult due to the nonlinear nature of the switching device. Therefore accurate compensation for this has not been presented up to now.

This paper will extensively deal with this problem and focuses on both the full analysis and the compensation of the distorted voltage to eliminate zero current clamping. In this compensation strategy, the gating signal is appropriately modified as the function of the current magnitude to synthesize accurate output voltage. Combining the dead time compensation method in ⁽¹⁾ which compensates the voltage distortion due to the voltage drop effect, the turn-on/turn-off time effect and the dead time effect, this compensation strategy can accurately compensate the voltage distortion even when the current is low. Moreover this scheme can be implemented by modifying only the software. So, with this method the reference voltage can be used as the inverter output voltage without any extra hardware. The method is suitable to apply space vector PWM that can synthesize accurate output voltage⁽⁷⁾. By applying this method to the general purpose PWM inverter system, a more accurate output voltage synthesis is possible. Experimental results are also presented to demonstrate the validity of the proposed method.

2. Analysis of Distortion Voltage

The reason to distort output voltage is the non-ideal characteristic of the diode. Assume that a-phase current in Fig. 1 is positive and nearly zero. Initially, the lower switch is assumed to conduct. In this period, the a-phase current flows

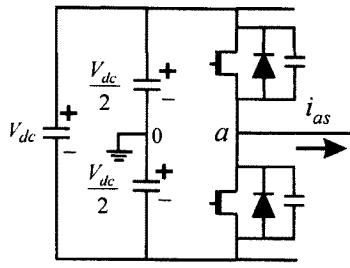
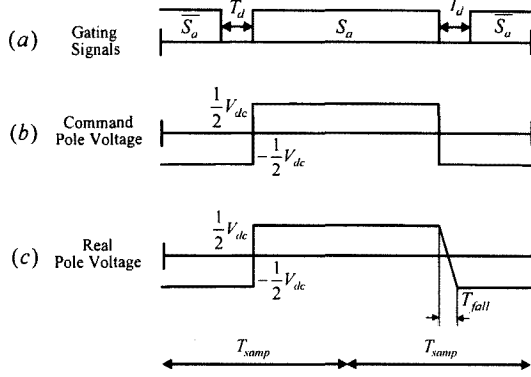
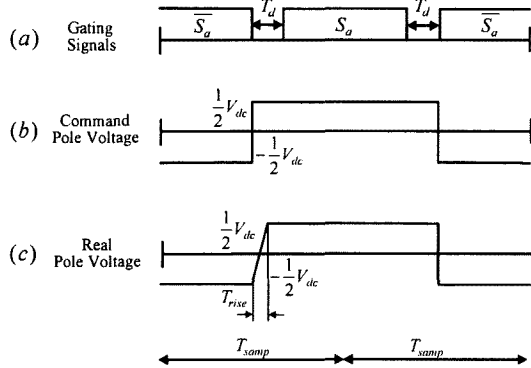
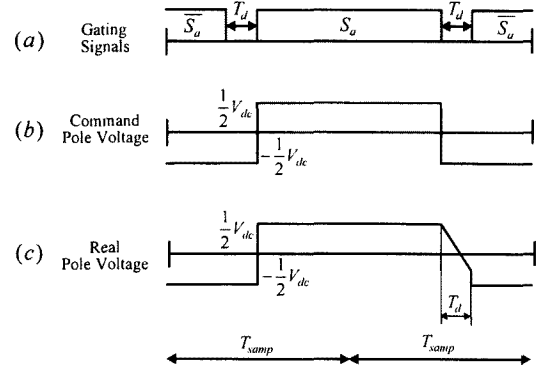
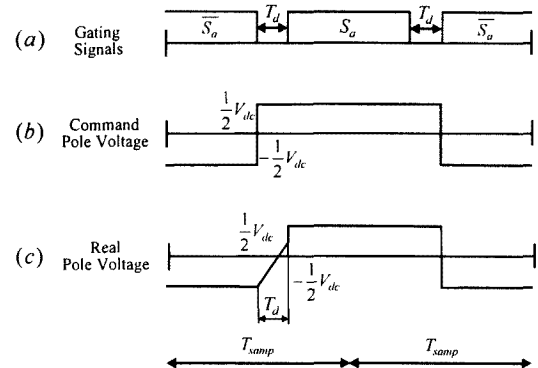


Fig. 1. One arm of the PWM Inverter


 Fig. 2. Voltage distortion ($i_{as} > 0$)

 Fig. 3. Voltage distortion ($i_{as} < 0$)

through the lower diode. To turn on the upper switch, first, the lower gating signal should be turned off before the upper gating signal is turned on for preventing the shoot-through phenomena. During dead time, the a-phase current continues to flow through the lower diode, thus the turn-off of the lower FET does not affect the current path. When the dead time expires and the upper gating signal is turned on, the upper FET is activated. This FET provides a low resistance path between the positive dc link and the point 'a'. The parasitic capacitor of the lower FET is charged almost immediately and the pole voltage goes high. For the upper FET turn-on, the a-phase current flows through the upper FET. To turn on the lower switch, the dead time is inserted between the upper gating turn-off and the lower gating turn-on. When these happen, the upper FET cannot support the current path. Also, the lower diode is reverse biased so that it will not provide the necessary current path at once. Notice

that only the lower diode can provide a current path. However, if it did so immediately, that would cause an instantaneous dc link voltage to appear across the upper parasitic capacitor before the lower diode can conduct. Therefore, the a-phase current must first charge the upper parasitic capacitor before the lower diode can conduct. Since the load is usually inductive and the inductive load acts as a current source for a short time, the voltage across the upper capacitor begins to rise linearly. The rate of change of the voltage is inversely proportional to the parasitic capacitance and directly proportional to the current. Thus, the actual pole voltage starts to decrease linearly as shown in Fig. 2. If the a-phase current is negative and nearly zero, the similar analysis can be possible and the real pole voltage linearly increases after the lower gating is turned off. This is depicted in Fig. 3. In low current region, this phenomenon is more remarkable, which results in the output voltage distortion. Therefore this leads to the degradation of the overall system performance in this operating region.


 Fig. 4. Voltage distortion ($i_{as} > 0, i_{as} \approx 0$)

 Fig. 5. Voltage distortion ($i_{as} < 0, i_{as} \approx 0$)

If the falling time of the pole voltage (T_{fall}) in Fig. 2 is greater than the dead time (T_d), this phenomenon looks somewhat different. If the dead time ends before the falling pole voltage reaches $-\frac{1}{2}V_{dc}$ as shown in Fig. 4, the turn-on of the lower FET makes the upper parasitic capacitor be charged immediately since the lower FET provides a low resistance path between the negative dc link and the point 'a'. Therefore, the pole voltage suddenly goes low as depicted in Fig. 4. In the case that the rising time of the pole voltage (T_{rise}) in Fig. 3

is greater than the dead time, the same analysis can be performed and the pole voltage is shown in Fig. 5. By this reason, though the rate of change of the pole voltage is directly proportional to the current, the voltage distortion has the nonlinear characteristic.

3. Equivalent Circuit

From above discussion, it is observed that the voltage distortion due to low magnitude current inevitably occurs when FET providing a phase current path is turned off. There are two voltage distortion modes. One occurs when upper FET conveying a positive and nearly zero phase current is turned off (positive current mode) and the other occurs when lower FET conveying a negative and nearly zero phase current is turned off (negative current mode). To investigate this phenomenon more intensively, the equivalent circuit during the voltage distortion mode will be examined.

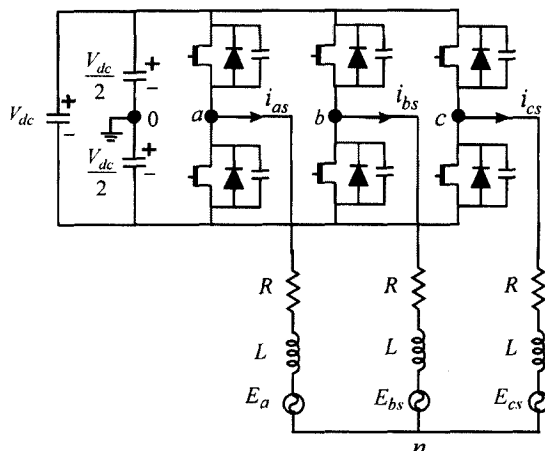


Fig. 6. 3-phase PWM inverter

The commonly used 3-phase PWM inverter is shown in Fig. 6, where the machine is modeled as an R-L network and a back-emf. During the a-phase dead time, the pole voltage V_{b0} and V_{c0} are defined by their switching functions and given by

$$V_{b0} = V_{dc} \left(S_b - \frac{1}{2} \right) \quad (1)$$

$$V_{c0} = V_{dc} \left(S_c - \frac{1}{2} \right).$$

To solve V_{a0} , consider the following equations.

$$V_{a0} = V_{an} + V_{n0} \quad (2)$$

$$V_{b0} = V_{bn} + V_{n0}$$

$$V_{c0} = V_{cn} + V_{n0}$$

The absence of a neutral connection in the motor forces the constraint which is

$$i_{as} + i_{bs} + i_{cs} = 0. \quad (3)$$

For any balanced load, the line to neutral voltages are constrained such that

$$V_{an} + V_{bn} + V_{cn} = 0. \quad (4)$$

These give

$$V_{n0} = \frac{1}{3} (V_{a0} + V_{b0} + V_{c0}). \quad (5)$$

By combining the voltage equations, V_{a0} is solved as

$$V_{a0} = \frac{3}{2} V_{an} + \frac{1}{2} (V_{b0} + V_{c0}) \quad (6)$$

$$= \frac{3}{2} V_{an} + \frac{1}{2} V_{dc} (S_b + S_c - 1).$$

During the voltage distortion mode, the pole voltage V_{a0} is, for the positive current mode,

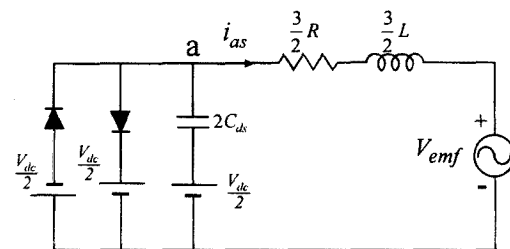
$$V_{a0} = \frac{V_{dc}}{2} - \frac{1}{2C_{ds}} \int i_{as} dt \quad (7)$$

and, for the negative current mode,

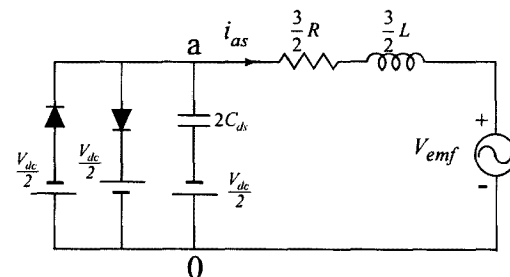
$$V_{a0} = -\frac{V_{dc}}{2} - \frac{1}{2C_{ds}} \int i_{as} dt \quad (8)$$

where the initial value of the integral is assumed to be zero and the capacitor C_{ds} denotes the parasitic capacitor between drain and source of the FET. The equivalent circuit is depicted in Fig. 7, where V_{emf} is defined by

$$V_{emf} = \frac{3}{2} E_{as} + \frac{V_{dc}}{2} (S_b + S_c - 1). \quad (9)$$

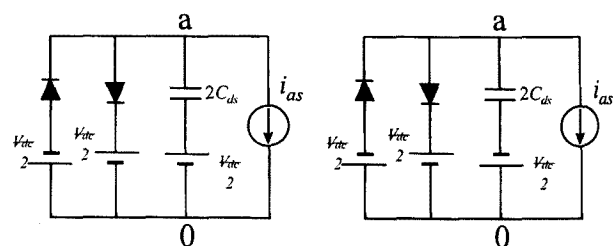


(a) positive current mode



(b) negative current mode

Fig. 7. Equivalent circuit



(a) positive current mode (b) negative current mode

Fig. 8. Equivalent circuit with current source

From Fig. 7, the voltage distortion phenomenon due to the parasitic capacitor seems to be complicated, that is, this phenomenon can be understood as the result of the second order electrical system. But, if the inductance L is sufficiently

large to keep the phase current to be nearly constant during the dead time, the inductor L can be considered as current source and the equivalent circuit can be represented as Fig. 8.

With this assumption, the transition of the pole voltage may linearly increase or decrease. Generally, this assumption is valid when the synchronous machine or the induction machine with the high stator transient inductance is used. In the case of the induction machine with the low stator transient inductance, this assumption is not valid and the voltage distortion phenomenon becomes complicated.

4. Compensation of Distortion Voltage

The mean reference pole voltage during the a sampling period is represented by

$$\overline{V_{a0}^*} = \frac{1}{T_{\text{samp}}} \int_0^{T_{\text{samp}}} V_{a0}^*(t) dt. \quad (10)$$

Similarly, the mean pole voltage during a sampling period is derived as

$$\overline{V_{a0}} = \frac{1}{T_{\text{samp}}} \int_0^{T_{\text{samp}}} V_{a0}(t) dt. \quad (11)$$

To compensate the distortion voltage, the mean pole voltage reference and the mean pole voltage during a sampling period must be same. To simplify the problem, the switch is assumed to have no turn-on and turn-off time and no voltage drop. Moreover, assume that a-phase current is

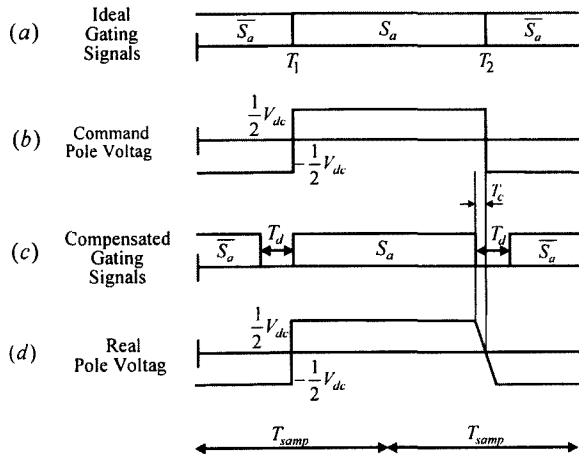


Fig. 9. Voltage distortion compensation ($i_{as} > 0$)

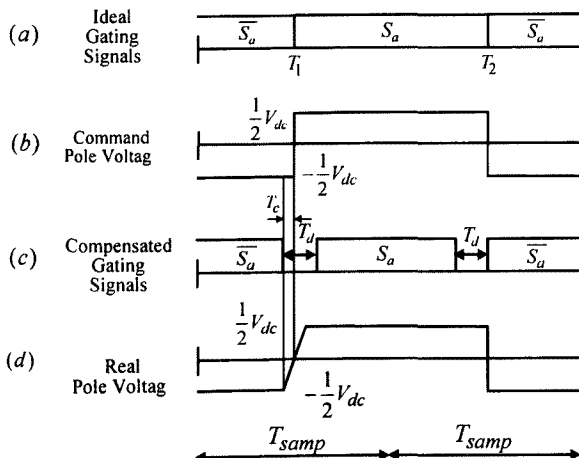


Fig. 10. Voltage distortion compensation ($i_{as} < 0$)

positive. If the switching pattern is given as shown in Fig. 9(a) to generate pole voltage as Fig. 9(b), the compensated gating signals such as Fig. 9(c) synthesizes the pole voltage in Fig. 9(d). At the first period, since the pole voltage goes high immediately as mentioned above, the turn-off of the lower gating at $T_1 - T_d$ guarantees an accurate voltage synthesis. However, at the last period, since the pole voltage slowly decreases, the upper gating should be turned off at $T_2 - T_c$. This makes the mean pole voltage reference and the mean pole voltage during a sampling period be equal. The time ' T_c ' denotes a compensation time. Though the falling time of the pole voltage is greater than the dead time, a suitable compensation time can reject the voltage distortion. In the case of the negative current, a similar compensation strategy can be applied and the voltage distortion can be eliminated as shown in Fig. 10. In short, the voltage distortion can be removed by the modifying the gating signal.

The compensation time is the function of the current magnitude flown through the power device. Moreover, this time is also related to the DC link voltage because the analogous action of the switching device during dead time interval is basically originated from the charging and discharging action of the junction capacitance between the drain and the source of the device. So, if the effective junction capacitance and the actual link voltage for charging or discharging action is closely known, on-line calculation for the voltage distortion can be accomplished according to the current level. Since, however, the factors for prediction are ambiguous and the relation between the factors has highly non-linearity, it is very difficult to compensate the voltage distortion caused by undesired switching action.

It is found during experimental test that the distortion phenomenon has different appearance according to the each switching device from other manufacturer. Fortunately, when one kind of devices from certain manufacturer is applied to inverter, the distortion level is nearly constant under same condition. So, this distortion can be almost removed by including the pre-calculated compensation time to the dead time compensation algorithm.

To find the compensation time, the off-line tests at some current levels are performed. When certain line current flows in one inverter arm under 150[V] DC link voltage condition, the mean time between the gating signal applying instance and actual switching instance is directly measured through the pole voltage traces captured in oscilloscope. This mean time can be used as the compensation time ' T_c '. Fig.11 shows one relationship between the current magnitude and the

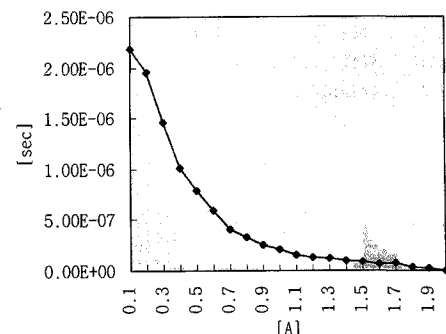


Fig.11. The measured compensation time versus device current.

measured compensation time of the switching device used in experiment. Of course, with the other kind of switching devices, the traces of the compensation time versus current level will differ from the presented in Fig.11 under same voltage condition. However, it can be verified through experiment that the outlines of the traces from other devices are roughly identical to that of Fig.11.

When the current magnitude is very small, the compensation time is nearly the dead time. Note that the compensation time can not exceed the dead time. In this study, $2.8[\mu\text{sec}]$ is selected for actual dead time. In the low current case, the proposed method produces the similar pole voltage generated without dead time compensation. So, in very low current magnitude region, no dead time compensation synthesizes a relatively accurate output voltage. The proposed compensation algorithm can be established in the conventional dead time compensator in which the gating signals are directly produced by software method⁽¹⁾. In this system, when the gating time calculated by software is loaded to the PWM device constructed with hardware ASIC(Application-Specific Integrated Circuit), the pre-setted dead time is automatically inserted to the actual gating pattern.

As previously mentioned, the voltage distortion compensation can be accomplished by considering the compensation time at dead time compensation instance. The compensation routine for one inverter arm is summarized again as follows. ' T_i ' and ' T_g ' mean the calculated gating time value for ideal pole voltage and the loaded time value for PWM ASIC, respectively.

if current > 0.	
if upper switch on	: $T_g = T_i - T_d$
else (lower on case)	: $T_g = T_i - T_c$
else (current < 0. case)	
if upper switch on	: $T_g = T_i - T_c$
else (lower on case)	: $T_g = T_i - T_d$

Of course, according to the current level, the compensation time should be calculated from look-up table in each sampling time. In fact, there are the turn-on/off time and voltage drop of switching in the PWM inverter and also these cause the voltage distortion. Thus these should be taken into account and can be minimized by the modifying the gating signal⁽¹⁾.

5. Experimental Verification

Experimental results are presented to show the validity of the proposed distorted voltage compensation strategy in the zero current clamping region. The power device is an FET whose current rating is 18[A] and voltage rating is 400[V] and the dc-link voltage is about 150[V]. The switching frequency is 8.33[kHz] and the sampling time is $60[\mu\text{sec}]$.

The voltage across the a-phase lower switch on the point of switching is shown in Fig. 12 when the a-phase current is positive and its magnitude is low and Fig. 13 when the a-phase current is positive and its magnitude is very low. In Fig. 12(c), it is observed that the voltage transition is somewhat slow, which results in the voltage distortion. In Fig. 13(c), the voltage transition is very slow and, as a result,

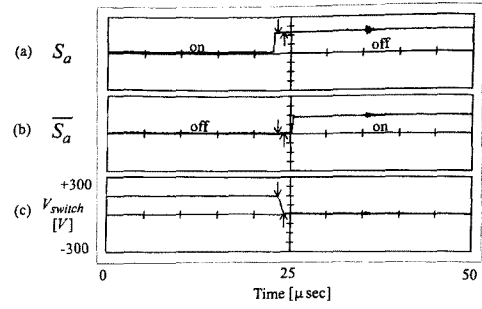


Fig. 12. Voltage distortion when the magnitude of the phase current is low

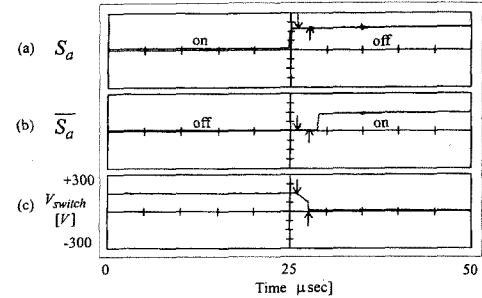


Fig. 13. Voltage distortion when the magnitude of the phase current is very low

the voltage rapidly goes low at the end of dead time. In this experiment, the voltage distortion by the zero current clamping is conspicuous when the magnitude of the current is below 2[A].

The proposed compensation method is compared with both the scheme in (1) and the no dead time compensation case. The comparisons are done to observe the current distortion when the constant phase voltage is applied to the induction machine whose transient inductance is about 7[mH]. The reference phase voltage is 5Hz, 10V in Fig. 14 and 20Hz, 10V in Fig. 15. The case of 5Hz, 10V is the relatively high current magnitude case and the case of 20Hz, 10V is the relatively low current magnitude case.

In the case of the high current magnitude, the current waveforms are shown in Fig. 14. The current waveform in the case of no dead time compensation (Fig. 14(a)) is much distorted and its magnitude is reduced due to the dead time effect. The current waveform in the case of scheme (1) (Fig. 14(b)) maintains sinusoidal waveform except zero current clamping region. The current waveform in the case of proposed scheme (Fig. 14(c)) maintains nearly sinusoidal waveform.

In the case of the low current magnitude, the current waveforms are shown in Fig. 15. The current waveform in the case of no dead time compensation (Fig. 15(a)) shows better waveform than in the case of scheme (1) (Fig. 15(b)). This is because, as mentioned above, no dead time compensation synthesizes a relatively accurate output voltage in the low current magnitude region. The current waveform in the case of proposed scheme (Fig. 15(c)) shows the best output voltage synthesis capability.

In these experiments, it is observed that the proposed scheme produces sinusoidal waveforms irrespective of current magnitude, thus generates an accurate output voltage. The improvement is apparent.

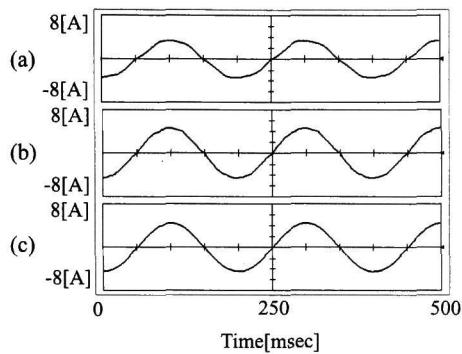


Fig. 14. Phase current waveforms in the case of the high current magnitude

[(a) no dead time compensation, (b) scheme (I), (c) proposed scheme]

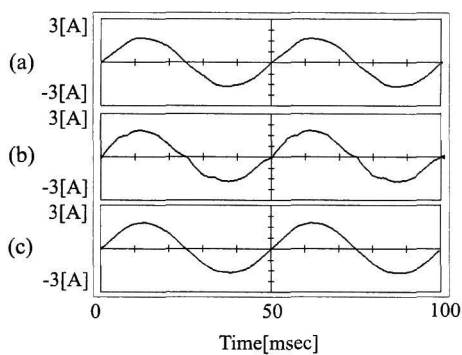


Fig. 15. Phase current waveforms in the case of the low current magnitude

[(a) no dead time compensation, (b) scheme (I), (c) proposed scheme]

6. Conclusions

In this paper, the detailed analysis of the zero current clamping phenomenon by the parasitic capacitor has been carried out. From this analysis, a novel distorted voltage compensation method which eliminates zero current clamping has been presented. With this method, the reference voltage can be used as the inverter output voltage without any extra hardware. By applying this method to the general purpose PWM inverter system, a more accurate output voltage synthesis is possible with no additional cost. Experimental results demonstrated the effectiveness of the proposed method.

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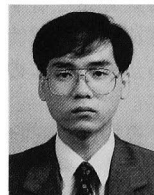
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